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Fiber-tip photonic crystal for real-time referenced biosensing in serum

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ABSTRACT

Fiber optic sensors have become increasingly well-established due to the many advantages they provide such as immunity to electromagnetic interference, multiplexing capabilities, and remote sensing. The coupling of light with a transducer at the tip of the optical fiber enables the detection of physical and biological parameters. 2D photonic crystals (PhC) can be designed to feature guided-mode resonances (GMR) characterized by a strong electric field at the PhC surface, providing a suitable tool for the detection of local refractive index variations (e.g. biomolecule adsorption). Here, we demonstrate the use of a PhC transferred to the tip of a single-mode fiber for biosensing. The control of surface chemistry provides a sensitive platform for the molecular recognition of antibody biomarkers. By integrating the fiber in a continuous flow platform, the real-time detection of anti-IgG in undiluted serum was achieved, with a limit of detection down to 60 pM. Moreover, the use of a reference channel is demonstrated to correct for signal drifts in real-time due to changes in bulk refractive index. These referenced fiber-tip PhC biosensors may pave the way for fluidic integrated systems in environmental, industrial, and healthcare applications, and open up the possibility of biosensing in the human body by integrating them into catheters.

1. Introduction

The development of biosensing techniques has progressed considerably in recent years, yet requirements in terms of rapid, efficient, and label-free detection of biomolecules persist. These needs have to be combined with practical considerations, enabling compatibility with on-site measurement (i.e. point-of-care technologies), requiring, for instance, the production of lightweight and low-cost sensors that can also operate in complex environments such as biological fluids.

In this respect, biosensors based on optical fibers are promising candidates, since they enable the delivery and collection of light through one-and-the-same fiber, are immune to electromagnetic interference, benefit from low manufacturing costs, and can provide multiparameter and remote sensing. They also display a range of advantages that make them highly attractive in the biomedical field (Socorro-et al., 2019): on the one hand, their silica structure provides a biocompatible material interface; on the other hand, their reduced size means they can be very minimally or non-invasive and allow for in-situ measurements by e.g. incorporating them in catheters (Carotenuto et al., 2018). Finally, the robustness of the materials enables them to operate in complex environments such as biological fluids.

To this end, over the last few decades, many efforts have been made

to develop fiber-optic biosensors (Tabassum and Kumar, 2020; Ochoa et al., 2021). As an example, fiber optic sensors based on surface plasmon resonance (SPR) are already commercially available, offering label-free sensing and real-time measurement with a good signal-to-noise ratio. (<https://foxbiosystems.com/>) On the other hand, the use of photonic structures offers the advantage of highly tailorable optical properties, while presenting less optical losses than SPR devices. In addition, these structures offer the ability to perform multiplexing, via wavelength division multiplexing and spatial multiplexing through the miniaturized fabrication of multiples structures on a single sensor. Importantly, the cleaved facet at the fiber tip provides an ideal platform for the integration of photonic devices and efficient light coupling directly into these structures. In many cases, the technologies employed rely on the detection of spectral changes in the reflected spectrum from photonic structures such as 1D and 2D photonic crystals (Singh et al., 2023). These techniques, based on the measurement of resonance shifts, offer considerable advantages in terms of signal accuracy and tolerance to intensity fluctuations, in contrast to intensity-based measurements (Socorro-et al., 2019). Several studies have demonstrated the use of fiber-tip sensors for biomarker detection aimed at diagnosing cancers (Bekmurzayeva et al., 2021; Loyez et al., 2021; Sypabekova et al., 2022; Yildizhan et al., 2023; Li et al., 2023), sepsis (Jeong et al., 2013; Wang

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et al., 2017), or cardiovascular events (Bonnez et al., 2023; Ayupova et al., 2021).

Although the high performance of these sensors enables them to detect biomolecules at sub-nanomolar concentrations, most of these systems are only compatible with controlled buffers that do not represent biological fluids (Inan et al., 2017). This limitation stems from the inability to combine detection of ultra-low concentrations with stable signal levels over timescales sufficient to ensure molecular recognition (typically of the order of tens of minutes). By using a reference technique, resonance shifts due to external factors such as changes in the bulk refractive index or temperature can be corrected. Recently, the utilization of a secondary channel (Velázquez-González et al., 2017) or a self-referencing probe (Janik et al., 2021; Bello et al., 2022) has been demonstrated, indicating that these systems, capable of compensating for external fluctuations, can achieve an accuracy more than ten times greater than that of sensors without compensation (Cano-Velázquez et al., 2023). Such self-referencing has so far been implemented in buffer solutions to compensate for bulk index and temperature variations. However, their transition to relevant conditions are essential for application at the point-of-care, i.e. referenced sensors that operate directly in undiluted complex media like serum or plasma are urgently needed.

Here, we develop a biosensor based on a photonic crystal that is placed on the tip of a fiber which can be easily integrated and functionalized with bioreceptors to detect biomarkers in complex biological fluids such as serum. 2D photonic crystals are mechanically transferred to a fiber tip enabling efficient coupling of light to the photonic crystal. The proposed configuration enables the simultaneous interrogation of two probes in a fluidic system, facilitating the correction of background signals due to changes in bulk refractive index, non-specific interactions, and temperature. In this way, the detection of anti-immunoglobulin antibodies with a limit of detection (LoD) of 60 pM was achieved directly in undiluted serum. We further demonstrate the ability of real-time detection in complex media, enabling the quantification of binding kinetics directly in biological fluids. We foresee that the presented approach will enable the label-free detection of biologically active molecules not only for healthcare but also for environmental monitoring in polluted water.

2. Design and characterization of the sensor

We employed numerical simulations to optimize the PhC design focusing on (i) a polarization independent resonance in the telecommunications wavelength range, warranting compatibility with commercial fiber-optic interrogation systems, and (ii) an optimized refractive-index sensitivity near the surface of the sensor for biosensing applications. A hexagonal PhC lattice displaying a polarization-degenerate mode was employed to provide a polarization-independent peak in the reflection spectrum (Cano-Velázquez et al., 2023; Conteduca et al., 2021). Simulations based on the finite element method (FEM) were performed to determine the optimum design of the PhC in terms of hole diameter and lattice constant. The position of the resonance, the sensitivity to a change in the bulk refractive index, and the quality factor were simulated (Figs. S1 and S2). As most of the electromagnetic field is confined within the holes, we find that the bulk index sensitivity of the PhC increases with larger diameter holes. However, increasing hole sizes also results in more efficient radiation into free space, thereby decreasing the quality factor (Grepstad et al., 2013). A trade-off must therefore be found to achieve optimal performance for biosensing. A hexagonal structure with lattice constants in the range 760 nm–800 nm and hole diameters between 110 nm and 140 nm was therefore selected. This results in a resonance wavelength in the range of 1560–1580 nm, with quality factors Q around 150 and a bulk refractive index sensitivity of 110–120 nm/RIU (RIU: refractive index units).

The PhCs fiber-tip sensors were then fabricated and transferred as previously described (Picelli et al., 2020). In short, the process is as follows: the structures were patterned on 250 nm thick InP membranes

using electron beam lithography followed by (inductively coupled plasma) reactive ion etching, and finally a large hole is etched in the substrate. A thin layer of silicon nitride of around 15 nm was deposited on the surface of the PhCs as a protective coating. The PhCs were then mechanically transferred onto the tip of a single-mode optical fiber in a process described earlier (Picelli et al., 2020). To this aim, a cleaved optical fiber is placed beneath the suspended membrane using a movable stage and aligned on a camera. The facet of the fiber is then brought into contact with the suspended PhC, which detaches by rupture of the four lateral supports (see Fig. 1a). The nanofabrication of the PhC cavities and the fiber-tip transfer are highly reproducible in subsequent fabrications. The lattice parameter is controlled with an accuracy of a few nm and the hole diameter with an accuracy of $\approx 5\%$, resulting in a reproducibility of the resonant wavelength within 10 nm. The wafer-to-fiber transfer process has shown a success rate of about 90% over >500 transfers performed over many years.

Reflection spectra were acquired using a four-channel fiber optic interrogator (Hyperion si255, Luna), based on a swept wavelength laser source with an output power of 0.25 mW (Micron Optics Inc. User Guide). The spectra were normalized by a reference spectrum measured by replacing the fiber-tip sensor with a fiber optic retroreflector (P5-SMF28ER-P01-1, Thorlabs). The recorded reflection spectrum (Fig. 1b) shows a Lorentzian lineshape centered at 1505 nm with a reflectance modulation of 68%, a full-width half-maximum FWHM = 20 nm, and a quality factor $Q = 77$. Slight deviations from simulations are observed, characterized by a shorter resonance wavelength and a lower quality factor compared to simulations. The differences observed are attributed to the manufacturing process of the crystal, which results in slightly larger holes than in the design. Moreover, the Q -factor is influenced by additional factors including surface roughness and hole conicity resulting from the etching step. As expected from the simulations, the hexagonal lattice structure of the PhC results in a polarization-independent response (Conteduca et al., 2021).

To measure a change in refractive index at the surface of the sensors in real-time, a setup equipped with a microfluidic channel was implemented in such a way that the spectra of two PhCs could be measured simultaneously. A 4-way fluidic chamber was used to accommodate the two PhC fiber-tip sensors while the remaining two inlets were used for the liquid flow. One of the two fiber tips will be used for the biosensing assay and will be referred to as "sensor", while the other will be referred to as "reference". Two syringe pumps were computer-controlled and linked to a mixing chip to dispense the liquid into the fluidic channel (Fig. 1c). The photonic crystal fiber-tip sensors were cleaned with an air plasma for 30 s to remove contamination, before introduction in the fluidic chamber. They were then equilibrated in phosphate buffer saline (PBS) for about an hour to stabilize the resonance wavelength due to temperature fluctuations before starting the assay. The flow rate was set at 50 $\mu\text{L}/\text{min}$, to ensure the stability of the sample throughout the entire measurement duration. This stability is demonstrated by the removal of the sensors from the fluidic chamber and their ability to be reused multiple times while maintaining their resonance and linewidth. The resonance wavelength of each fiber-tip PhC sensor was tracked in real time by fitting a Lorentzian function around the central wavelength.

3. Results

3.1. Bulk sensitivity

We first characterize the bulk sensitivity of the PhC, which refers to the resonance shift for a given change in bulk refractive index. As mentioned above, two fiber-tip PhCs can be simultaneously measured within the setup. Hence, two PhCs with the following designs were characterized: (sensor: $a = 800$ nm, $r/a = 0.16$; reference $a = 790$ nm, $r/a = 0.16$). The sensitivity measurement is performed by running a series of solutions with different refractive indices through the flow chamber. The solutions used are mixtures of ethanol (EtOH) and water at different

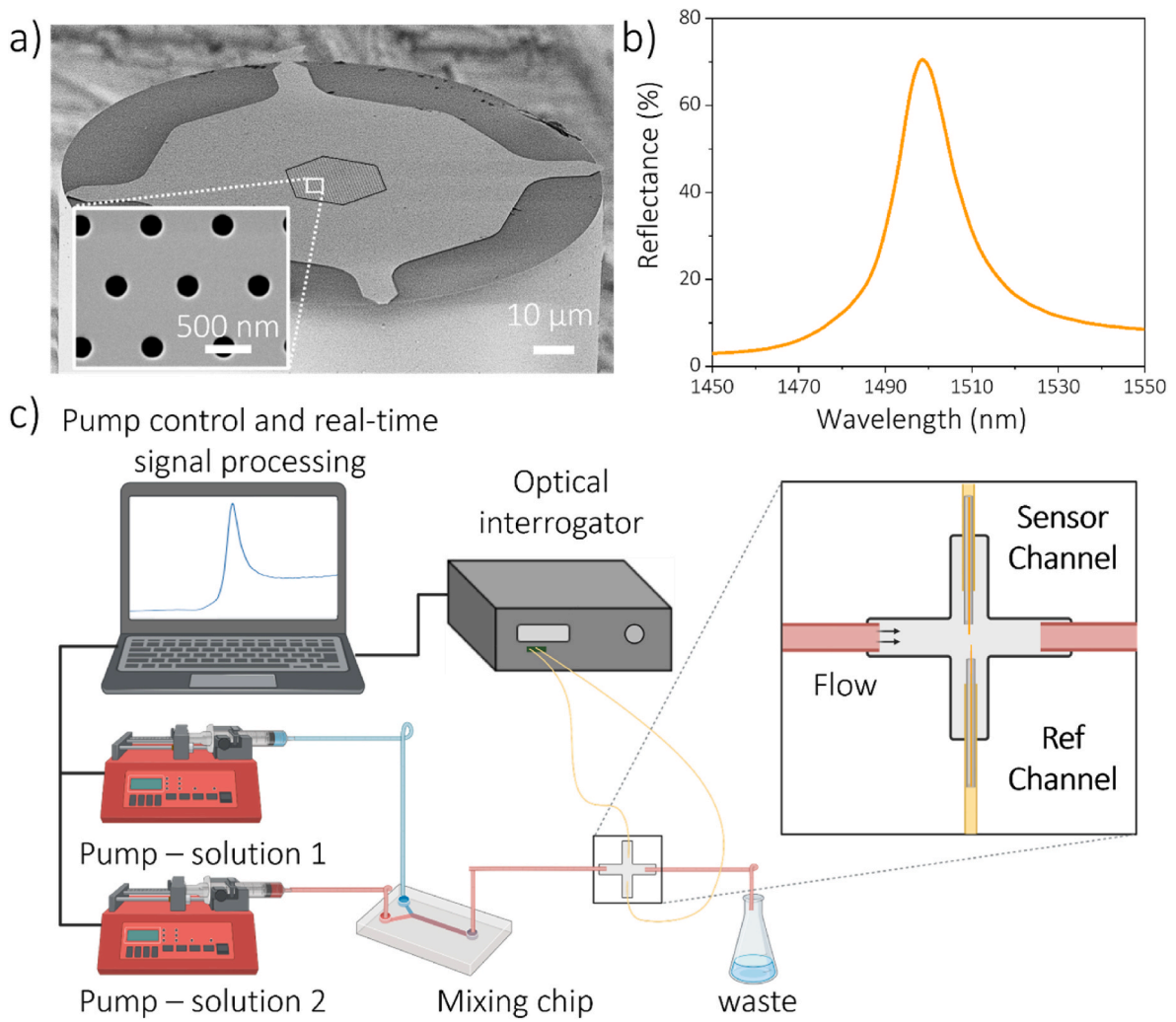


Fig. 1. a) Scanning electron microscope (SEM) image of the PhC fiber-tip sensor, the inset corresponds to a close-up of the central part of the PhC showing the hexagonal lattice. b) Typical reflection spectrum obtained for a hexagonal PhC fiber-tip sensor. c) Schematic representation of the setup capable of accommodating two sensors in a microfluidic cell. The flow is controlled by a dual-pump system and a mixing chip. Solution 1 and Solution 2 refer to water and ethanol, respectively for the bulk refractive index measurement and to PBS and solution containing the target for the assays. The sensors are illuminated and measured by an optical interrogator.

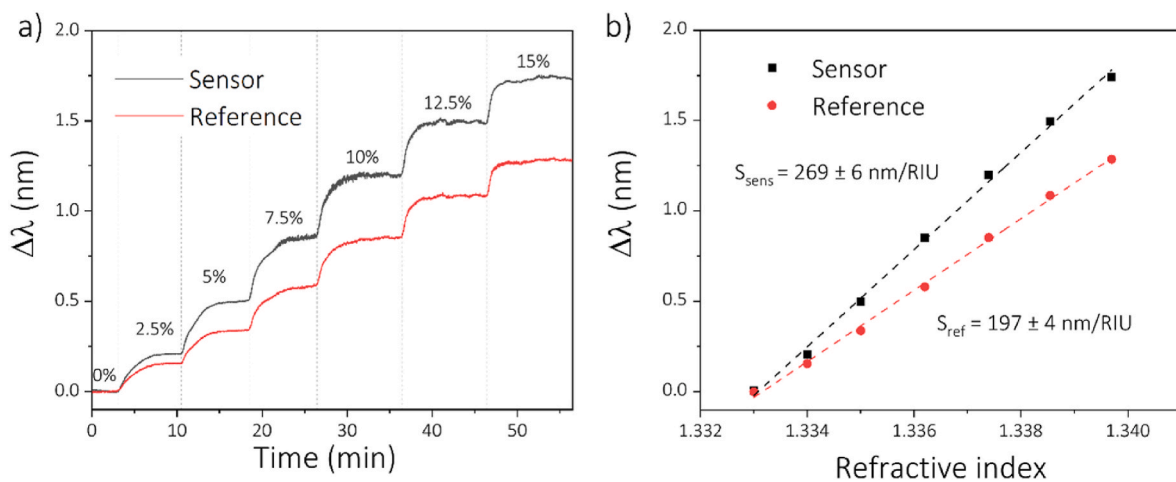


Fig. 2. a) Shift in resonance wavelength as a function of time at different mixing ratios of ethanol in water. b) Resonance shift as a function of the refractive index. The dashed lines indicate linear fits to the data points yielding the bulk index sensitivity S .

ratios (from 0% to 15% in volume fraction, corresponding to a refractive index of 1.3333–1.340 at 1550 nm, as measured with a commercial refractometer at 590 nm and corrected for wavelength) (Ferreira et al., 2018). The EtOH/water solutions were then infused stepwise (flow rate of 50 $\mu\text{L}/\text{min}$), gradually increasing the ethanol content by steps of 2.5%, during which the resonance peak was fitted every second (Fig. 2a).

After each injection, a rapid red shift in the resonance indicated a change in the bulk refractive index. The shift in resonance wavelength for each solution was then plotted as a function of the refractive index (Fig. 2b). The data were fitted by a linear curve resulting in sensitivities of $269 \pm 6 \text{ nm}/\text{RIU}$ and $197 \pm 4 \text{ nm}/\text{RIU}$ for the sensor and reference, respectively. The difference in the bulk sensitivity between the two PhCs comes from the slight difference in lattice constant as listed above. According to the simulations (see Fig. S1), the bulk sensitivity is expected to be higher with a larger lattice constant, which is indeed observed. However, a slightly lower sensitivity value was expected from the simulations and this variation could be attributed to the larger holes of the PhC during the manufacturing process, as explained before. These variations in the experimental refractive index sensitivities are often seen, which is most likely due to the reproducibility of the hole diameter of the PhC and the transfer. Furthermore, the sensitivity can be affected by the surface state and possibly due to the liquid not completely filling the holes or creeping between the fiber and the PhC. The bulk sensitivities found are within the typical range of sensors based on photonic crystals found in the literature (Pitruzzello and Krauss, 2018).

3.2. Functionalization and anti-body detection in buffer

A first bioassay was then performed in a PBS buffer solution, in order

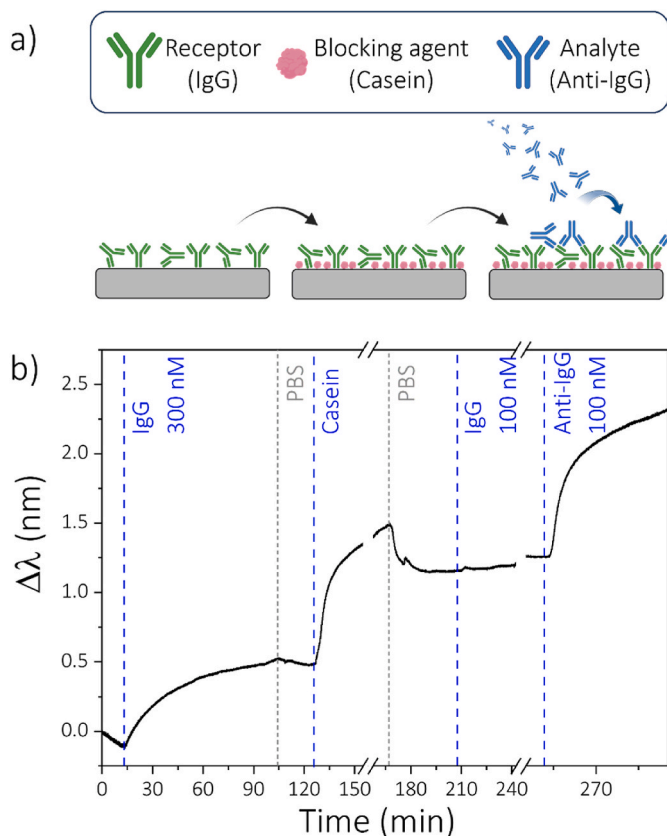


Fig. 3. a) Schematic illustration of the functionalization of the PhC surface and the bio-assay for the detection of anti-IgG. b) The wavelength shifts of the sensor during the entire bio-assay with each step indicated by the vertical dashed lines.

to test the sensitivity and specificity of the sensor. The samples were functionalized with IgG capture probes to facilitate the detection of anti-immunoglobulin (anti-IgG) antibodies. The functionalization process (Fig. 3a) involved the physisorption of IgG onto the plasma-cleaned PhC surface, followed by the adsorption of casein, which acts as a blocking agent to reduce non-specific interactions. Fig. 3b shows the resonance shifts in real-time during the functionalization and sensing steps for one of the PhCs. After IgG injection at 300 nM, a red shift of about 0.5 nm is observed, indicating the physisorption of antibodies onto the PhC surface within 30 min. When rinsed with PBS, the resonance remained stable, demonstrating that no detectable desorption of IgG occurs. The casein step was performed by injecting a 10 mg/mL casein buffer and also showed a rapid red shift indicating the adsorption of casein on sites not covered by IgG. After rinsing, the casein is partially removed indicated by the blue shift of the resonance, accounted for by the removal of weakly bound molecules from the surface. The signal then stabilizes rapidly after the injection of PBS.

The specificity of the sensor was first evaluated by introducing a non-binding antibody, for which the same IgG antibody is used as for the functionalization step. IgG at 100 nM was injected for 40 min, resulting in a very weak red shift of the resonance (about 20 pm). The sensor was then exposed to anti-IgG at the same concentration. The sensor response to anti-IgG showed a rapid response, resulting in a total shift of around 1 nm. These experiments demonstrate the specific detection of anti-IgG with negligible non-specific interactions by IgG.

3.3. Antibody detection in serum

To evaluate the performance of the biosensor in complex media, experiments were performed in serum. The sensor was immersed in an Eppendorf filled with undiluted fetal bovine serum (FBS) and left for 2 h to stabilize the signal while stirring. The sequential injection of FBS spiked with anti-IgG at different concentrations was realized to determine the thermodynamic parameters and to evaluate the dynamic range of our biosensor. Fig. 4a shows the response of the sensor for concentrations of anti-IgG ranging from 100 pM to 100 nM. The sensor shows a red shift of the resonance that increases to 3 nm for increasing concentration. The shifts were extracted for each concentration after 40 min of injection and the resulting dose-response curve was fitted with a Langmuir model (eq. S(1), Fig. 4b). Langmuir model extraction yielded a dissociation constant (K_D) of 29 nM, consistent with the value reported in the literature for this specific antibody pair (Sinibaldi et al., 2018). It is important to note that equilibrium is not reached after 40 min for all concentrations, which we will discuss further below.

Even at the highest concentrations, saturation of the sensor is not reached, and, surprisingly, adsorption kinetics appear slower for these higher concentrations. This contradicts the prediction of the Langmuir model where adsorption kinetics scale with the association rate and the concentration of ligand. This behavior was further studied by plotting the initial rate, at which the resonance shifts, as a function of ligand concentration. The rate was obtained from the slope of a linear fit to the curve directly after injection. We find that the initial rate increases with anti-IgG concentration up to 10 nM, after which it decreases (Fig. 4b).

These discrepancies with the standard Langmuir model can be explained by considering a more complex interaction in which recognition between receptor and ligand is heterogeneous. By considering a model composed of two independent interactions (known as the Langmuir dual-site model) (Albers et al., 2011), it becomes easier to understand this behavior. Whereas a 1:1 receptor-ligand model shows only one type of interaction with a single well-defined rate constant, a dual-site model assumes two types of binding sites each with their own kinetic parameters. As a result, one binding mode with high affinity dominates at low concentrations, while at higher concentrations, a second binding mode with lower affinity dominates the total response (Karlsson et al., 2006). This type of behavior explains the resonance shifts and initial rates we observe (see Fig. S4).

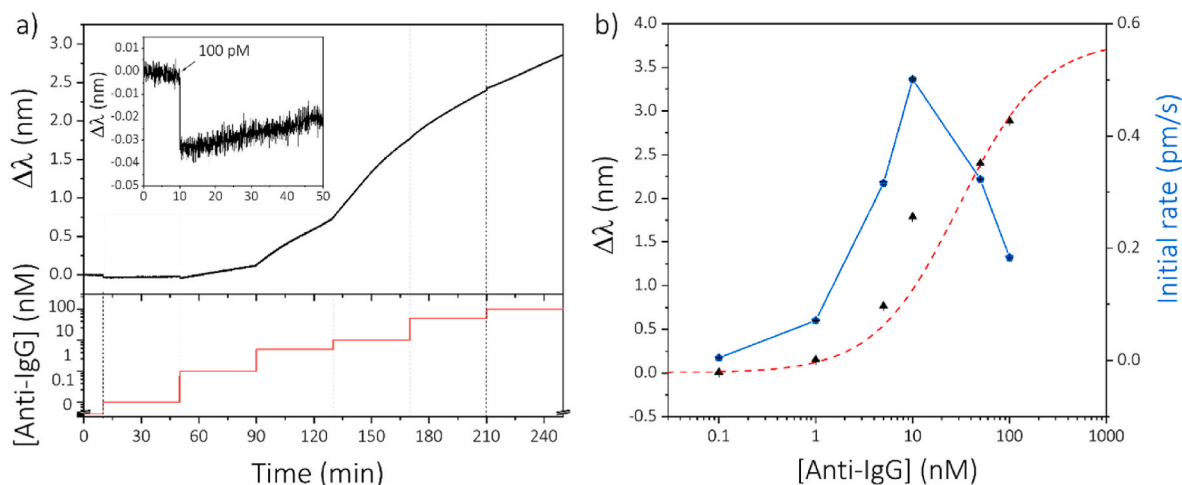


Fig. 4. a) Wavelength shift of the PhC sensor in blood serum after injection of Anti-IgG with different concentrations. The dotted lines serve as a visual guide for each concentration increase. Inset: zoom-in of the 100 pM injection b) Dose-response curve of the wavelength shift (black dot) and initial rate (blue dots) for each concentration. The dashed red lines correspond to the fit with the single-analyte Langmuir model. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

To directly confirm this dual-site adsorption we performed measurements in buffer over much longer injection periods at a single concentration (Fig. S3). Indeed, after a rapid response in the first few minutes, a second slower response was observed over the 3.5 h injection period, corresponding to the second binding mode with a lower association rate (and thus a lower affinity). The use of a single exponential function results in an inadequate fit, whereas the use of a double exponential function (eq. S(2)) provides a robust fit, supporting the presence of multiple components in the adsorption process. This type of two-phase adsorption is commonly observed for polyclonal antibodies (Albers et al., 2011) and reflects different affinities within the sample. The geometrical characteristics of our system may suggest that adsorption processes differ between the PhC surface and the holes inside the structure.

To evaluate the LoD we determine the standard deviation of the resonance wavelength as $\sigma_\lambda = 2.44$ pm, by measuring the signal for 10 min (corresponding to 300 spectra) before the first injection of anti-IgG. Considering this value as the noise level and by extracting the sensitivity of the spectral response to a change in concentration ($d\lambda/dC$, corresponding to the slope of the linear part of the dose-response curve), the LoD of the system is estimated as $LoD = 3 \sigma_\lambda / (d\lambda/dC) = 60$ pM (9 ng/mL). The performance of the sensor is therefore comparable with similar fiber-tip biosensor technologies in terms of detection limits (Pitruzzello and Krauss, 2018). It is noteworthy to point out that the LoD (17 pM for the detection of CD44 protein (Bekmurzayeva, 2021), 50 pM for HER2 protein (Loyez et al., 2021), and sub-pM for interleukin IL-8 (Rashidova et al.)) found are in controlled buffers, as no studies on fiber-tip sensors show results in complex environments to our knowledge.

3.4. Real-time referencing in serum

Due to its high sensitivity, the sensor response is affected by the background drift due to temperature, non-specific interactions, and changes in the bulk refractive index of the surrounding medium. This is particularly the case for sensing applications in serum where the bulk refractive index is higher than commonly used buffers due to the presence of a large pool of proteins and nutrients. Here we integrate a second PhC that acts as a reference to correct for these non-specific signals. This reference also overcomes the variability encountered during the sensor fabrication step, as the calibration measurement carried out beforehand defines the intrinsic sensitivity of each sensor (cf. Fig. 2). We test the ability of this dual-channel sensor to correct for the background signals by monitoring its response in real-time during a drastic change of the

medium, namely from pure buffer ($n_{PBS} = 1.3342$) to ligand-spiked serum ($n_{FBS} = 1.3417$).

To enable real-time referencing and extraction of the signals due to ligand binding, the reference PhC was introduced into the fluidic chamber after sensor functionalization with IgG and thus only exposed to casein, providing a fully blocked surface. Fig. 5a shows the resonance shift for both the sensor and the reference PhC. We observe a strong shift of the sensor PhC due to the physisorption of IgG capture probes, followed by a shift due to casein blocking that is similar for both PhCs with a shift of about 1 nm. After rinsing and stabilizing, FBS spiked with anti-IgG at 200 nM was injected resulting in a large red-shift of 1.42 and 1.85 nm for the reference and the sensor, respectively. These shifts reflect the contributions of the change in the bulk refractive index, temperature, non-specific interactions, and the specific binding of anti-IgG to IgG in the case of the sensor.

It is evident that the primary contribution to the signal arises from changes in the bulk refractive index, as indicated by the sensitivity measurements (see Fig. 2b). Indeed, the refractive index variation observed between PBS and FBS ($\Delta n = 0.0075$, measured before experiment with a refractometer) would correspond to a red shift of 1.5–2 nm, depending on the exact bulk index sensitivity of the PhC. On the other hand, the temperature sensitivity is typically in the range of a few tens of pm/K (Cano-Velázquez et al., 2023), with a specific value of 61 pm/K recorded for this specific sensor (see Fig. S5). Based on this value, it is evident that temperature drifts cannot account for the observed wavelength shifts. Quantification of the shift due to non-specific interactions is achieved by monitoring the signal after rinsing with PBS buffer. Following rinsing, the sensor retains a shift of approximately 1 nm, while the reference almost returns to its initial position, exhibiting a final shift of 0.10 nm (see Fig. S6). This indicates that the response of the sensor remains stable despite the presence of interfering molecules with high concentrations in a complex environment. By monitoring the reference signal throughout the assay, the effect of bulk RI and non-specific interactions can be effectively compensated for.

The biosensor signal was corrected in real-time by subtracting the sensor signal from the reference, while correcting for the slightly different bulk index sensitivities of the PhCs:

$$\Delta\lambda_{corr} = \Delta\lambda_{sens} - \frac{S_{sens}}{S_{ref}} \Delta\lambda_{ref}.$$

Herein S_{sens} and S_{ref} are the bulk sensitivities of the sensor and reference, respectively (see Fig. 2b). The corrected signal $\Delta\lambda_{corr}$ is then extracted in real-time (Fig. 5b). The signal shows a rapid red-shift within

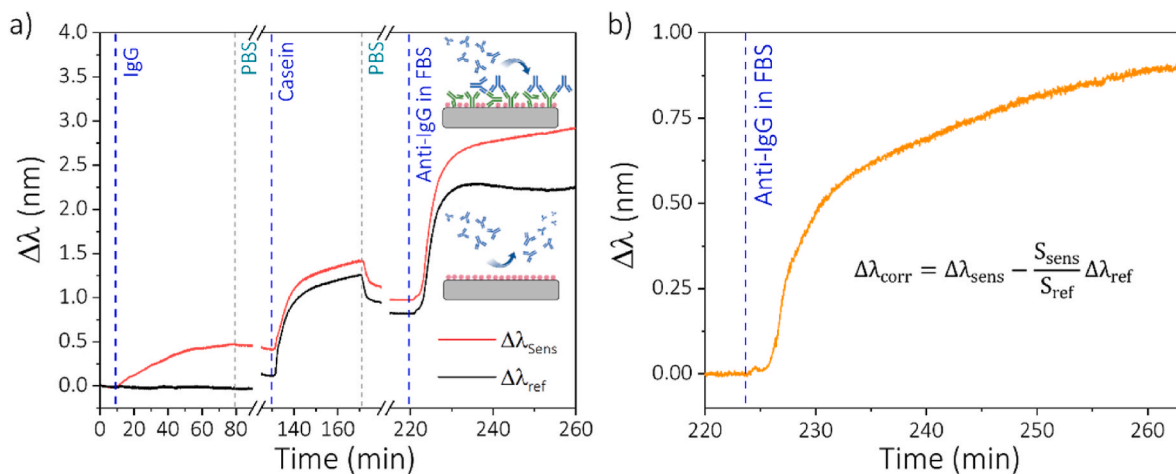


Fig. 5. a) Wavelength shift during the bio-assay with the sensor (in black) and the reference (in red). b) Corrected signal for the bulk refractive index during the injection of Anti-IgG at 200 nM in serum. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

5 min followed by a slower increase without reaching saturation after 30 min. This kinetics is similar to the one observed in buffered conditions with a similar shift (about 1 nm).

Signal referencing also improves stability over extended measurement durations. Signal drift, inherent in all measurements, can occasionally constrain the detection of extremely low concentrations with slow kinetics. For instance, in the case of the 100 pM measurement, a shift of 11 pm occurred within 40 min, where signal drift may become predominant (see inset in Fig. 4a). Signal drift was evaluated for the sensor both before and after signal correction (corresponding to Fig. 5a and b, respectively). Prior to referencing, drift was measured over a 30-min period before FBS injection on several samples (see example Fig. S7). Drift was reduced by almost a factor of 2 (from an average of 260 ± 75 to 160 ± 100 pm/h depending on the sample), demonstrating the ability of dual detection to compensate for non specific signals. The referencing along two lines provides clear advantages regarding the correction for non-specific interactions and bulk refractive index changes as well as a nearly factor 2 for the drift reduction in static conditions. By mitigating signal drift, we effectively minimize fluctuations and uncertainties, ensuring a more consistent and reliable measurement environment. Ultimately, by tackling signal drift, we increase the reliability and accuracy of our analysis techniques, paving the way for accurate detection not only in biological fluids but also in other media, such as water, where environmental monitoring and analysis are crucial.

4. Conclusion

In this study, we demonstrated experimentally the use of a fiber-tip photonic crystal with a tailored design for relevant analyte biosensing. The implementation of the sensor and the simplicity of the readout allowed for the real-time monitoring of the functionalization steps as well as for the specific detection of anti-IgG in the nanomolar range. The system showed a LoD of 60 pM in undiluted blood serum. The use of a second sensor acting as a reference enabled the correction of external factors such as the bulk refractive index and showed potential for live referencing. The innovative design and flexibility of these biosensors present a promising avenue for real-time, in-line detection alongside catheter applications, offering efficient monitoring solutions in medical contexts. Moreover, their adaptability extends beyond healthcare, with the control of the surface chemistry, potentially enabling the sensitive detection of water pollutants, and thereby contributing to environmental monitoring efforts. With their multifaceted capabilities, these sensors represent a significant advancement in both medical and environmental fields, demonstrating their versatility and potential for

addressing a variety of detection challenges with precision and efficiency.

CRedit authorship contribution statement

Mathias Dolci: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Paco Dreverman:** Methodology, Investigation, Data curation. **Mildred S. Cano-Velázquez:** Software, Resources, Data curation. **Arthur L. Hendriks:** Methodology, Investigation, Data curation. **Emiel Veth:** Data curation. **P.J. van Veldhoven:** Methodology. **Andrea Fiore:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Peter Zijlstra:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biosx.2024.100573>.

Data availability

Data will be made available on request.

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